

|   |   |                                                                                                                                                                                                                          |
|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   | $v_f = 3.0 \text{ m/s}$                                                                                                                                                                                                  |
| 4 | B | <p>Since no net external force acts on the system of 2 particles, final momentum = initial momentum</p> <p>Considering vertical momentum,</p> $p_x \sin\theta + (-p_y \sin\alpha) = 0$ $p_x \sin\theta = p_y \sin\alpha$ |
| 5 | D | Let $l$ be the distance from hip joint pivot to centre of mass                                                                                                                                                           |